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RESEARCH ARTICLE

Transfer Learning and Custom Loss Applied to Transformer-Based Text Translation for Sign Language Animated Subtitles

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ABSTRACT Online learning's rise presents unique challenges for the deaf community, particularly in understanding educational videos. This research addresses the problem by proposing a solution to generate animated subtitles in the Indonesian Sign Language System, namely SIBI (Sistem Isyarat Bahasa Indonesia). The existing method produces word-by-word animations from directly recognized spoken text that are too fast and difficult to follow. We developed a text translation model for the Moodle application to shorten the original spoken text of the educational videos into SIBI text without losing the crucial information. We propose transfer learning to train our transformer-based models and use a custom loss function to ensure alignment with the SIBI dictionary. Pre-trained models mBART50 and NLLB200 were fine-tuned on the SIBIVID-MP12 dataset, which was created in collaboration with Special Education teachers. Experiment results show that the proposed method improves translation metrics significantly, with the best performance observed for the NLLB200 model fine-tuned with our proposed custom loss, achieving sacreBLEU, chrF++, METEOR, and ROUGE-L improvements of 71%, 9.79%, 22.92%, and 14.55%, respectively. This research demonstrates the potential for enhanced inclusivity in online learning for the deaf community.

INDEX TERMS Machine translation, mBART50, NLLB200, SIBI, subtitle generation, transfer learning, custom loss.

I. INTRODUCTION

In this era of rapid technological progress, online learning is developing and becoming increasingly popular. Schools and universities are starting to adopt and adapt to online learning systems which are carried out both synchronously and asynchronously. People with special needs, including the deaf, often face challenges in utilizing online learning. Deaf people have difficulty receiving information from learning video media. This happens because video media displays information through images and sound, while deaf people have difficulty hearing these sounds [1], [2]. In addition, low language skills make it difficult for deaf people to speak reading fast subtitle text on videos, thus hindering their understanding of the material presented through fast-paced text [2]. According to the World Federation of the Deaf, there are over 200 sign languages used around the world and 70 million deaf people using them [3]. In Indonesia, there are two recognized sign languages: SIBI (Sistem Isyarat Bahasa Indonesia = Sign Systems for Indonesian Language) and BISINDO (Bahasa Isyarat Indonesia = Indonesian Sign Language). BISINDO is informal, whereas SIBI is more formal and is used in schools. This research focuses on SIBI.

Several SIBI studies have been conducted to develop more inclusive solutions. The process of creating SIBI gesture animations has previously been carried out in research at

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the Faculty of Computer Science, University of Indonesia (Fasilkom UI) conducted by Wihalim and Khairevy [4] and Permadi [5]. Suggestions from the functionality test and user evaluation carried out by Permadi show that the SIBI sign language subtitle animation generated by the Moodle application based on the video subtitle text word for word causes the animation to move too fast, because the animation follows word for word spoken by the speaker. Additionally, if a word is not available in the SIBI dictionary, the model spells the word letter by letter, which accelerates the animation speed even further and makes it increasingly difficult for viewers to follow.

To solve the fast-moving animation, our research aims to develop a machine translation (MT) model on learning video transcript texts that will be used to generate SIBI sign language animated subtitles [6]. The MT converts the input text taken from the video voice recordings into output SIBI text used to generate the sign language animations. SIBI text is a more concise version of standard text, optimized to use words available in the SIBI dictionary whenever possible. This conciseness facilitates the generation of more streamlined animations. We utilize two Transformer-based base models, namely mBART50 [7] and NLLB200 [8], as the baseline models. Our preliminary experiments found that both models have been proven effective in MT tasks for Indonesian language. The baseline models are then fine-tuned to our domain task, SIBI text translation.

We propose transfer learning approach by fine-tuning the model using additional custom loss function. The objective of this custom loss function is to minimize the frequency of spelling out words not present in the SIBI dictionary by penalizing token predictions that are outside the dictionary. This custom loss functions involves applying additional penalties to token predictions that are not listed in the SIBI dictionary. This dictionary includes a collection of basic words as well as combinations of various affixes in Indonesian. Through the implementation of this custom loss function, every token prediction that is not in the dictionary will be subject to a penalty. The aim of this strategy is to encourage the model to more frequently select tokens that suit predetermined linguistic and contextual needs, thereby increasing the overall relevance and accuracy of the model output.

Our research applies two ways to penalize the models using our proposed custom loss function: Loss Integration Type 1 and Loss Integration Type 2. The first variant adds the custom loss directly to the total loss, thereby enhancing the sensitivity of the model to the custom loss by emphasizing the importance of SIBI dictionary adherence. In this approach, the custom loss is fully integrated into the total loss without any scaling, ensuring that the model gives significant attention to minimizing this custom loss. The second variant, however, combines the model's standard loss and the custom loss proportionally using an alpha factor. This approach allows the loss to be adjusted flexibly, making it proportional and enabling a balance between the impact of the standard loss and the custom loss. Various weight and penalty values are experimented with to find the best model performance.

Current approaches to sign language processing are often confined to specific disciplines, leading to an incomplete understanding of the problem. For instance, many computer science studies focus on developing algorithms for recognizing or translating signed content, but these efforts often exclude Deaf individuals with firsthand experience and lack linguistic insights necessary for handling the complexities of sign languages. Furthermore, these algorithms are typically trained on datasets that fail to represent real-world usage, limiting their practical application [9], [10], [11].

To address these issues, an interdisciplinary approach is necessary. This involves integrating deafness studies to understand the community's needs, linguistics to identify the structural elements of sign languages, Natural Language Processing (NLP) and machine translation for effective modeling and translation, computer vision for detecting signed content, and Human-Computer Interaction (HCI) for designing user-friendly systems that serve the community effectively [9].

In our research, we developed a new dataset named SIBIVID-MP12 to support these interdisciplinary efforts. The dataset includes videos across 12 school subjects, annotated with SIBI subtitles, and was created with input from SLB (Sekolah Luar Biasa = Special School for deaf and hard of hearing students) teachers to ensure it meets the specific needs of deaf students. Observations at Santi Rama School provided insights into sign language teaching methods. The dataset, comprising 441 rows, underwent rigorous preprocessing, including removing non-ASCII characters and ensuring alignment with the SIBI dictionary. The data was split into 80% for training and 20% for testing, further divided into validation and test sets to support robust algorithm development.

Our experiments use two fine-tuning variations for both baseline models. This variation consists of fine-tuning (FT) and fine-tuning by adding a custom loss function (FT+CL). Additionally, to integrate the custom loss with the total loss we apply two ways integration, type 1 and type 2 aforementioned. The experiments apply various fine-tuning parameters settings including random state 42 for reproducibility, 200 epochs, patience 10 to prevent over-fitting, max token length 243 to ensure data integrity, and batch size 1-4 for efficiency and memory. A custom loss function with logarithm penalty is applied to improve the prediction accuracy of words in the SIBI dictionary, maintain consistency, and ensure the relevance of model results in the context of sign languages. We also experiment with some penalty values. In total, we produced 106 different models based on these various settings.

The experimental results show that fine-tuning (FT) improves the performance of both mBART50 and NLLB200 models in translating Indonesian text into SIBI compared to the baseline model. Our proposed fine-tuning with custom

loss (FT+CL) can further improve the models' performances. Among various fine-tuning settings, the best model is NLLB200 FT+CL variant trained using batch size 4 and loss integration type 2 that integrates custom loss and standard loss using flexible weight adjustment with weight size of 0.2, and a penalty value of 0.1. Compared to the baseline model NLLB200, the best NLLN200 FT+CL model improves the translation performance with an increase in sacreBLEU value of 71%, chrF++ value of 9.79%, METEOR value of 22.92%, and ROUGE -L value of 14.55%.

To evaluate the whole SIBI sign language subtitle animation system applied to the educational video our survey team [12] conducted a user acceptance test and post-test involving respondents from senior high SLB students. The user acceptance test results show that the speed of the sign language animation generated using ground truth SIBI text is acceptable and helps the students understand the material better. This is also in line with the post-test results. Students watching videos with sign language subtitle animation achieve higher learning scores.

Overall, the contribution of this research can be summarized as follows:

1. Develop SIBI text translation module for SIBI Sign Language Subtitle Animations: The research enhances the generation of SIBI sign language subtitle animations for deaf students, solving the issue of fast-paced animations that hinder comprehension.
2. Propose Transfer Learning Approach using Fine-tuning of Transformer-based Models: By fine-tuning mBART50 and NLLB200 models, the study improves the translation accuracy of Indonesian text into SIBI, ensuring relevance and contextual accuracy.
3. Propose a Novel Custom Loss Function: This research introduces a novel custom loss function that penalizes token predictions not listed in the SIBI dictionary, enhancing the model's performance and translation accuracy.
4. Create a new dataset namely SIBIVID-MP12. We perform Comprehensive Data Collection and Validation: The research collects and prepares the specialized dataset through interviews and video annotations.

The remainder of this paper is organized as follows. Section II reviews work related to this research. Section III presents the proposed architecture of the model that is being used and its training stages. Section IV provides the implementation details of our SIBI translation models. Section V presents the experiment results and analysis, and Section VI concludes the research.

II. RELATED WORKS

A. SIGN LANGUAGE GENERATION

Sign language avatars, which are computer-generated animations of human figures, play a crucial role in making information accessible to individuals who are Deaf or Hard of Hearing (DHH) and prefer sign language. These avatars offer a valuable alternative to written text, especially for those with lower literacy levels in the written language, as sign languages are typically not written [13]. Unlike videos of human signers or artist-created animations, sign language avatars are particularly advantageous in scenarios where automatic content generation is necessary, such as on websites with frequently changing content [9].



FIGURE 1. Research pipeline used by Wihalim, Khairevy, and Permadi.

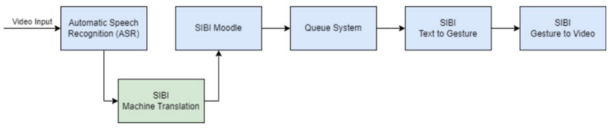


FIGURE 2. Proposed pipeline by adding SIBI machine translation between ASR module and SIBI Moodle.

The ability of these avatars to provide sign language translations in real-time or on-demand makes them a powerful tool in bridging the accessibility gap for the DHH community. However, while existing systems focus on well-known sign languages such as American Sign Language (ASL) or British Sign Language (BSL), there has been limited exploration into the automatic generation of Indonesian Sign Language (SIBI) content, which is vital for ensuring inclusivity within Indonesia [9].

In current approaches to sign language translation, the generation of avatars typically relies on a symbolic representation of the signed content, which is often prepared by a human author [14], [15], [16], [17], [18]. This method is prevalent in various systems, where the symbolic representation serves as a blueprint for generating the sign language animation. When integrated into a translation system, the process usually begins with an initial step that converts spoken or written language into a symbolic representation of the target sign language. This symbolic plan is essential, whether the content is human-authored or automatically translated, as it guides the creation of the sign-language message [15], [17], [19].

Despite the widespread use of symbolic representations, there is no universal standard for these symbolic plans. Various representations have been proposed, each with its unique characteristics and applications. The lack of a standardized approach can lead to inconsistencies in translation quality and the effectiveness of sign language communication across different systems [9]. Our research aims to address these challenges by developing a more consistent and accurate translation pipeline for Indonesian Sign Language (SIBI).

Previous research conducted at the Faculty of Computer Science, University of Indonesia (Fasilkom UI) by Wihalim and Khairevy [4] and later by Permadi [5] has explored the creation of SIBI gesture animations. The system developed in these studies involves extracting audio from videos and

converting it into text using an Automatic Speech Recognition (ASR) system with pipeline as presented in **FIGURE 1**.

This text is then used to generate 3D animated interpretations of sign language. However, these word-by-word animations, particularly when generated letter-by-letter words not in SIBI dictionary, tend to move too quickly, leading to difficulties for viewers in following the content. This problem is exacerbated in fast-paced videos, where the animations become increasingly hard to comprehend, limiting the effectiveness of the educational material for the deaf community.

Interviews with SLB (Sekolah Luar Biasa = Special School for deaf and hard of hearing students) teachers highlighted that in real interactions, sign language presenters do not replicate every word verbatim but instead focus on conveying the core message. This discrepancy between the generated animations and actual sign language communication suggests a need for improvement in the system to ensure it better serves the needs of the deaf community.

To address these issues, our research introduces a novel approach that enhances the existing system by integrating SIBI machine translation between ASR and SIBI Moodle as presented in **FIGURE 2**. Unlike the previous word-by-word animations generated directly from original spoken text sentences, our method focuses on translating the entire sentences from ASR into SIBI text sentences, the summarized version of the input sentences based on SIBI dictionary. Our proposed SIBI MT ensures that the essence of the original content is preserved in the translated SIBI text and the sign language subtitle animation is delivered at a pace that is easier for the audience to follow. A survey is also carried to validate user's acceptance of the whole application.

B. SIGN LANGUAGE TEXT TRANSLATION

In general machine translation (MT) methods can be classified into two categories: 1) Rule-Based and Statistical Methods (SMT) and 2) Neural Machine Translation (NMT). SMT methods rely on predefined linguistic rules and statistical models, respectively. SMT methods involve translating text by using statistical models derived from analyzing large amounts of parallel text data, where the probabilities of word sequences are calculated based on their occurrences in the training data [20], [21]. NMT is a modern translation technique that uses artificial neural networks to directly map sentences from one language to another. Unlike traditional methods such as SMT, which rely on separate components for different translation tasks, NMT models handle the entire process within a single, unified model. This approach allows NMT to produce more fluent and accurate translations by learning from large datasets. It is particularly effective for complex language pairs and has set a new standard in machine translation performance [22], [23].

Some NMT studies on sign language translation (SLT) propose an End-to-End SLT systems [24], [25]. This approach has shown to be particularly effective in scenarios with limited training data, though its performance improves as more data becomes available.

Another NMT study uses a Multimodal SLT systems to enhance translation accuracy by integrating multiple types of data alongside textual information [26]. By combining these different data modalities, these systems can better capture the context and meaning of the source material, which is crucial for accurately translating sign languages. This approach is particularly useful in low-resource settings and helps address challenges like translating rare or complex words.

In our case, to development SIBI text translation models we performed a comparative study on several NMT models, such as T5 [27], M2M100 [28], mBART50 [7], NLLB200 [8]. Text-to-Text Transfer Transformer (T5) model was introduced by Raffel et al. [27] to simplify various language problems by treating them as text-to-text tasks. This approach allows systematic comparisons between pre-training goals and transfer strategies across different NLP tasks. Experimental results demonstrated that the T5 model achieved state-of-the-art performance in various NLP benchmarks, especially when larger models are trained with sufficient data.

Fan et al. [28] introduced the M2M100 model, a many-to-many multilingual translation model supporting 100 languages without relying on English as an intermediary. This model employs a Transformer [29] architecture with dense scaling and language-specific sparse parameters. Experimental results showed that M2M100 achieved state-of-the-art performance in various multilingual translation tasks, while also reducing English-centric bias in translation systems.

Tang et al. proposed mBART50 [7], a multilingual sequence-to-sequence model that supports 50 languages, including Indonesian, making it an excellent choice for our task. The ability of both models to handle Indonesian natively is crucial, as it allows for more accurate and contextually relevant translations, which are essential for generating high-quality SIBI translations. Model mBART50 extends mBART25 [30], a sequence-to-sequence denoising autoencoder pre-trained multilingual on monolingual text. This model showed significant performance improvements in various MT tasks, especially for low-resource languages. The denoising techniques employed, such as text infilling and sentence permutation enabled the model to learn to reconstruct corrupted texts, thereby enhancing robustness and performance in translation tasks.

Finally, Team et al. proposed NLLB200 model [8], a significant achievement in the field of MT designed to support more than 200 languages, including low-resource languages. This model was developed by the NLLB team with the goal of reducing language barriers globally, targeting languages underserved by previous MT technologies. The NLLB200 provides a solution to the limitations of legacy MT models such as the M2M100, which, despite supporting multiple languages, still exhibited a performance gap between high- and low-resource languages.

Based on our preliminary experiment, the NLLB200 and mBART50 models were specifically chosen for this research due to their support for multiple languages, including Indonesian. NLLB200 is designed to handle low-resource

languages, making it well-suited for translating into SIBI, which has a relatively limited amount of annotated data available. Both models are used as our baseline models.

We employ transfer learning in this research due to its ability to leverage pre-existing knowledge from large, pre-trained models and adapt it to specific tasks, such as translating Indonesian text into SIBI. Given the relatively small size of domain-specific datasets like those available for SIBI, training models from scratch would be highly resource-intensive and less effective. By using transfer learning, we can fine-tune large, pre-trained models to our specific task, significantly reducing the amount of data and computational power required while improving the overall performance of the model in a specialized domain like sign language translation.

Moreover, we introduce a custom loss function, designed to ensure that the translation adheres closely to the SIBI dictionary, further refining the accuracy and relevance of the generated content. This approach not only addresses the issue of overly rapid animations but also aligns more closely with how sign language is naturally communicated, improving the overall learning experience for deaf users. By integrating these improvements, our solution significantly enhances the inclusivity and effectiveness of educational content delivered through platforms like Moodle for the deaf community.

Compared to previous related works, the novelty of our research lies in several key innovations. First, we introduce a custom loss function, which penalizes the model when it generates tokens that are not found in the SIBI dictionary. This approach ensures that the generated translations are more accurate and relevant to the linguistic needs of the SIBI language, addressing a significant gap in previous models that often relied on more generic translation techniques. Additionally, the research explores different methods of integrating this custom loss into the overall model training process, allowing for more flexible and effective fine-tuning. By focusing specifically on SIBI and addressing the unique challenges associated with translating into this sign language, our research offers a more tailored and effective solution than previous approaches, which often did not account for the specific needs of sign language translation.

III. METHODS

We propose a new pipeline application to generate SIBI sign language subtitle animations as presented in **FIGURE 2**. We incorporate the SIBI machine translation (see the green box) to translate the original voice text produced by the ASR module into SIBI text and then feed the text to SIBI Moodle. In this section, we describe the architecture of the proposed machine translation models and the training process. This includes the custom loss functions used to optimize performance for machine translation of sign language. Both models leverage Transformer-based architectures to handle the complexity of the task.

A. MODEL ARCHITECTURE

Our SIBI machine translation model takes an input text $\boldsymbol{X}$ where it is the original text and change it to sentence $\hat{\boldsymbol{Y}}$ where it is more compact text.

$$\hat{\boldsymbol{Y}} = \text{model}_\theta(\boldsymbol{X}), |\hat{\boldsymbol{Y}}| <= |\boldsymbol{X}|$$

Here $\boldsymbol{X}$ is the input text and $\hat{\boldsymbol{Y}}$ is the output text, $|.|$ is the number of words in the sentence, and model_θ is a model and the parameters that are being used.

The proposed model architecture leverages advanced Transformer-based neural networks, specifically the mBART50 and NLLB200 models, for machine translation tasks in sign language. Both models share the foundational Transformer architecture, which is effective in sequence-to-sequence tasks.

The Transformer model, introduced by Vaswani et al. [29], is designed for handling sequential data through self-attention mechanisms. This design enables parallel processing and captures long-range dependencies. The architecture consists of an encoder and a decoder, each comprising multiple layers of self-attention and feed-forward networks. (“Multibody Models Generated from Natural Language”)

The mBART50 [7] extends the mBART25 [30] model to support 50 languages via multilingual fine-tuning. The architecture features 12 layers each in the encoder and decoder, with a model dimension of 1024 and 16 attention heads. This results in approximately 680 million parameters. A distinctive feature of mBART50 is the use of an additional LayerNorm embedding, which stabilizes and improves the training process [7].

The NLLB200 incorporates the Sparsely Gated Mixture of Experts (MoE) mechanism. This mechanism facilitates efficient handling of translation tasks across 200 languages. The encoder includes MoE layers at specified intervals. Multiple experts, managed by a gating network, process tokens efficiently. The MoE approach optimizes computational resources by activating relevant parts of the network per task [8].

Both models have a similar data flow, with the main differences being the additional LayerNorm embedding in the mBART50's encoder and decoder stacks and the different training data used for the SentencePiece tokenizer [31] in each model. The data flow in both models begin with tokenized text inputs processed by the encoder, generating hidden state representations $\boldsymbol{H}$:

$$\boldsymbol{H} = \text{Encoder}(\boldsymbol{X}). \tag{1}$$

These hidden states are then used by the decoder along with the shifted right version of the target sequence $\boldsymbol{Y}_{\text{shifted}}$ to produce the output sequence $\hat{\boldsymbol{Y}}$.

$$\hat{\boldsymbol{Y}} = \text{Decoder}(\boldsymbol{H}, \boldsymbol{Y}_{\text{shifted}}) \tag{2}$$

The model then computes the output probabilities, selects the token with highest probability, and generates the predicted text. The loss is calculated as a combination of cross-entropy loss and the custom loss.

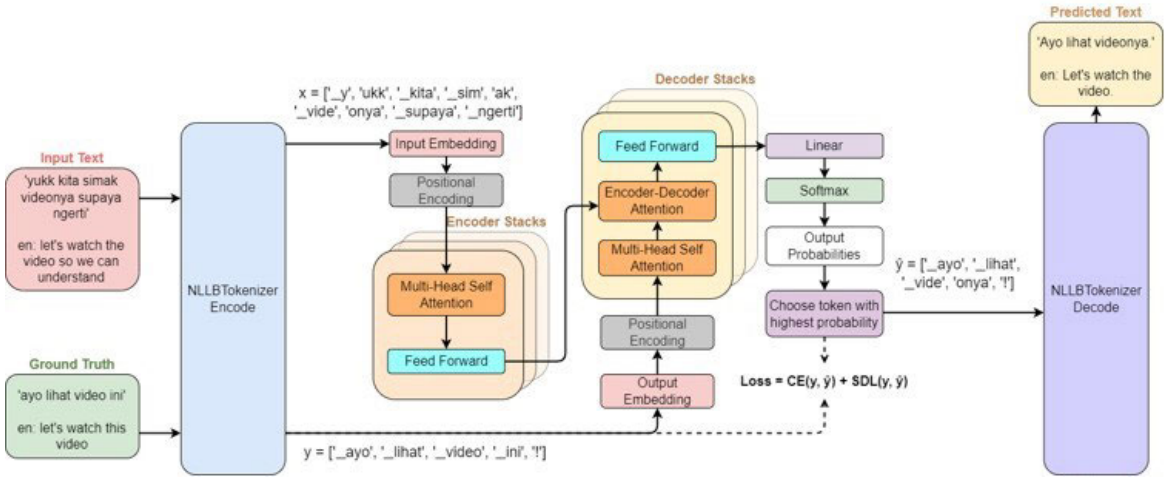


FIGURE 3. Simplified architecture of mBART50 model. Model translate the input text "yukk kita simak videonya supaya ngerti" (English: let's watch the video so we can understand) to the ground truth "ayo lihat video ini" (English: let's watch this video). mBART50, using MBARTTokenizer, includes an additional LayerNorm embedding in both the encoder and decoder, generating the predicted text "Ayo lihat videonya." (English: Let's watch the video.) with a loss computed as Cross Entropy (CE) and custom loss (SDL).

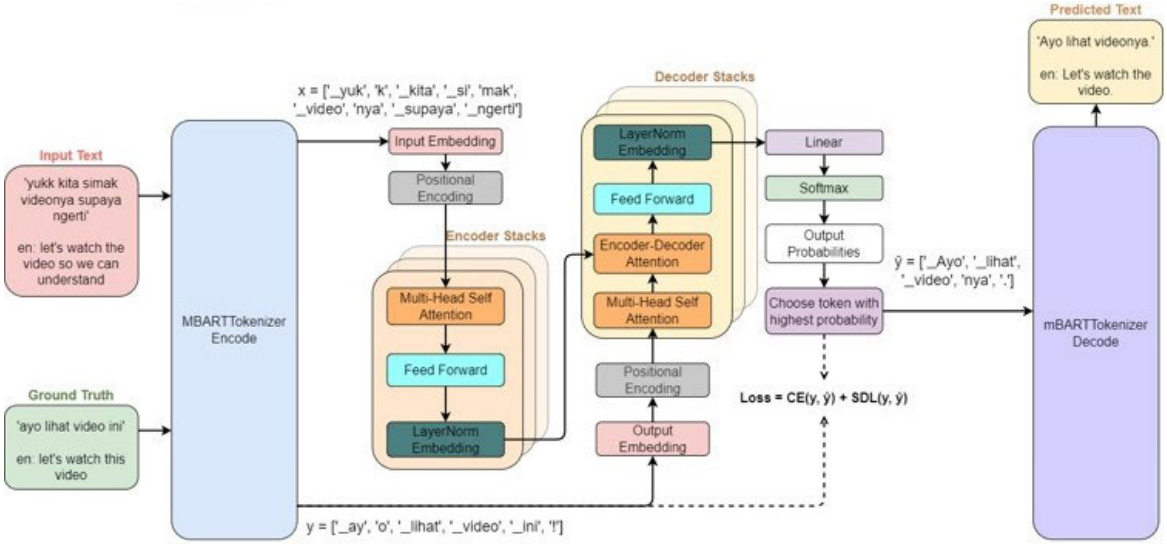


FIGURE 4. Simplified architecture of NLLB200 model. Model translate the input text "yukk kita simak videonya supaya ngerti" (English: let's watch the video so we can understand) to the ground truth "ayo lihat video ini" (English: let's watch this video). NLLB200 uses NLLBTokenizer, multi-head self-attention, and feed-forward networks in its encoder and decoder, generating the predicted text "ayo lihat videonya!" (English: let's watch the video!) with a loss computed as Cross Entropy (CE) and custom loss (SDL).

In the NLLB200 architecture, the input text "yuk kita simak videonya supaya ngerti" (English: Let's watch the video so we can understand) is tokenized and encoded by the NLLBTokenizer. The tokenized input, represented as $\boldsymbol{X} = [$"_y", "ukk", "_kita", "_sim", "ak", "_vide", "onya", "_supaya", "_ngerti"], passes through the input embedding and positional encoding layers before entering the encoder stacks. The encoder stacks consist of multi-head self-attention and feed-forward layers. The encoded representation $\boldsymbol{H}$ is then fed into the decoder stacks, which include multi-head self-attention, encoder-decoder attention, and feed-forward layers. The decoder outputs are passed through a linear layer and softmax to produce output probabilities. The predicted text is generated by selecting the token with the highest probability and decoded by the NLLBTokenizer.

In the mBART50 architecture, the input text "yuk kita simak videonya supaya ngerti" (English: Let's watch the video to understand) is tokenized and encoded by the MBARTTokenizer. The tokenized input, represented as $\boldsymbol{X} = [$"_yuk", "k", "_kita", "_si", "mak", "_video", "nya", "_supaya", "_ngerti"], passes through the input embedding, positional encoding, and LayerNorm embedding layers

before entering the encoder stacks. The encoder stacks consist of multi-head self-attention, feed-forward, and LayerNorm embedding layers. The encoded representation $\boldsymbol{H}$ is then fed into the decoder stacks, which include multi-head self-attention, encoder-decoder attention, feed-forward, and LayerNorm embedding layers. The decoder outputs are passed through a linear layer and softmax to produce output probabilities. The predicted text is generated by selecting the token with the highest probability and decoded by the MBARTTokenizer.

Despite both models using the SentencePiece tokenizer, the resulting tokenization differs due to the different training data used for each model's tokenizer. This results in different tokenization behavior and thus influences the models' translation performance.

FIGURE 3 and **FIGURE 4** illustrates simplified architectures of the mBART50 and NLLB200 models, respectively. These figures highlight the flow of data through the encoding and decoding stages.

B. MODEL TRAINING

The fine-tuning process involves adapting pre-trained models to the specific characteristics of the dataset. Two variations of fine-tuning were employed: standard fine-tuning and fine-tuning with a custom loss function.

1) STANDARD FINE-TUNING

Standard fine-tuning involves further training of the pre-trained models using the processed dataset. This allows the models to adapt to domain-specific data and improve their translation accuracy.

2) FINE-TUNING WITH CUSTOM LOSS FUNCTION

Fine-tuning with a custom loss function incorporates an additional custom loss function during training. This addresses specific dataset nuances. The custom loss function ensures that the models generate translations using only words present in the SIBI dictionary.

We propose two ways to integrate the models with custom loss functions: integration type 1 and integration type 2. The first variant directly adds the custom loss to the total loss. This integration method enhances the sensitivity of the model to custom loss, emphasizing the importance of the SIBI dictionary adherence. The equation for this variant is as follows:

$$\text{LOSS}_{\text{total}} = \text{LOSS}_{\text{standard}} + \lambda \text{LOSS}_{\text{custom}} \tag{3}$$

where $\text{LOSS}_{\text{total}}$ is the model total loss, $\text{LOSS}_{\text{standard}}$ is the standard MT loss, $\text{LOSS}_{\text{custom}}$ is proposed custom loss, and λ is the weight proportion value.

The second variant combines the standard loss and custom loss proportionally. This integration method allows a flexible adjustment between the standard and custom loss contributions, providing a balanced approach to training. The equation for this variant is:

$$\text{LOSS}_{\text{total}} = (1-\lambda)\,\text{LOSS}_{\text{standard}} + \lambda \text{LOSS}_{\text{custom}}, \tag{4}$$

TABLE 1. Description of the models and hyperparameters used in the empirical study. Baseline refers to pre-trained models without fine-tuning. FT refers to models fine-tuned on the dataset. FT+CL refers to models fine-tuned with a custom loss function.

Model	Batch size	Weight Variant	Penalty
mBART50 FT	1, 2, 3, 4	-	-
mBART50 FT+CL	1, 2, 3, 4	Integration Type 1: 0.01, 0.05, 0.1, 0.5, 1 Integration Type 2: 0.2, 0.4, 0.5, 0.6, 0.8	0.01, 0.05, 0.1, 0.5, 1
NLLB200 FT	1, 2, 3, 4	-	-
NLLB200 FT+CL	1, 2, 3, 4	Integration Type 1: 0.01, 0.05, 0.1, 0.5, 1 Integration Type 2: 0.2, 0.4, 0.5, 0.6, 0.8	0.01, 0.05, 0.1, 0.5, 1

where $\text{LOSS}_{\text{total}}$ is the model total loss, $\text{LOSS}_{\text{standard}}$ is the standard MT loss, $\text{LOSS}_{\text{custom}}$ is proposed custom loss, and λ is the weight proportion value.

The custom loss or $LOSS_{custom}$, is defined as:

$$\text{LOSS}_{\text{custom}} = \frac{1}{N}\sum_{i=1}^{N}\begin{cases} -\log(p) & \text{if } t_i \notin D \\ 0 & \text{if } t_i \in D, \end{cases} \tag{5}$$

where N is the number of tokens, t_i is the token at the i-th index, D is the SIBI dictionary, and p is a penalty constant. Each token t_i is checked against the dictionary D. If t_i is in the dictionary, its contribution to loss is 0. Otherwise, its contribution is $-\log(p)$. The total contribution of all tokens to the loss is averaged by dividing by the total number of tokens N.

For the penalty value p, we selected a moderate value of 0.1 along with values close to it, as shown in **TABLE 1**. This decision assumes that an excessively high penalty may negatively impact the model's learning process, while an overly small penalty may fail to produce a noticeable effect. Thus, 0.1 strikes a balance between these extremes, ensuring optimal performance without compromising the model's effectiveness.

We developed a total of 106 different models based on these settings, with 2 models serving as the baseline. The settings proposed include variations in batch size, loss function type, including the weighting type, the weights, and the penalty constant. This extensive variation helps in identifying the most effective training configuration. The results of this empirical study will be discussed in Section V.

IV. EXPERIMENTS

In this section, we detail the experimental setup used to evaluate the performance of our machine translation models. This

includes a description of the dataset, the experimental setup, and the evaluation metrics used to measure the effectiveness of our approach.

A. DATASET

The dataset utilized in this research is the SIBIVID-MP12, comprising 441 annotated text pairs derived from 12 educational YouTube videos. These annotations were meticulously crafted by special education teachers to ensure they align with the standards of the Indonesian Sign Language (SIBI). The data underwent a thorough cleaning process to enhance quality and consistency, which involved the removal of non-ASCII characters, extra whitespace, and corrections according to the SIBI dictionary. The language used in the dataset is Indonesian

The dataset was split into training, validation, and testing subsets following an 80:10:10 ratio. Specifically, 352 instances were used for training, 44 for validation, and 45 for testing. This split ensures that the model is evaluated on data it has not seen during training, thus providing a robust assessment of its performance.

B. EXPERIMENTAL SETUP AND DESIGN

This section focuses on experiment-specific parameters such as batch sizes, penalty values, and weight variants (detailed in **TABLE 1**). Metrics for model performance evaluation, including sacreBLEU, METEOR, chrF++, and ROUGE-L, are applied as described earlier. Specific preprocessing steps and model architecture details are outlined in the Methods section.

C. MODEL EVALUATION

The SIBI MT models were evaluated using several metrics to ensure a comprehensive assessment of their performance. The primary metrics used were sacreBLEU, METEOR, chrF++, and ROUGE-L.

SacreBLEU measures the accuracy of translated text by comparing it to a set of reference translations. This metric is designed to provide a single score that reflects how similar the machine-generated translation is to one or more human-generated reference translations. It accounts for n-gram precision and includes a brevity penalty to penalize short translations. SacreBLEU also ensures consistency by applying

its own metric-internal tokenization, thus standardizing the preprocessing step and eliminating variations caused by different tokenization schemes [32]. The formula for BLEU is as follows:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \tag{6}$$

where BP is the brevity penalty, w_n is the weight, and p_n is the precision n-grams [33].

METEOR is another metric used to evaluate machine translation quality. It considers both precision and recall, and it goes beyond exact word matches by incorporating synonymy and stemming. METEOR is particularly good at capturing the meaning of the translation rather than just surface-level word matches. The formula for METEOR is expressed as:

$$\text{METEOR} = \frac{10 \cdot P \cdot R}{R + 9 \cdot P} \cdot (1 - \text{Penalty}), \tag{7}$$

$$\text{Penalty} = 0.5 \cdot \left(\frac{C}{U}\right)^3, \tag{8}$$

where P is precision, R is recall, C is the number chunks, and U is the number of matches. METEOR is particularly useful for evaluating semantic similarity and capturing synonymy, which other metrics may overlook [34].

ChrF++ is a character-level metric that evaluates translation quality based on n-gram precision and recall. This metric is useful for capturing fine-grained details in the translation that word-level metrics might miss. It is particularly effective for languages with rich morphology. The chrF++ metric is computed as:

$$\text{chrF} = F_\beta = \left(1 + \beta^2\right) \frac{P \cdot R}{\beta^2 \cdot P + R}. \tag{9}$$

This formula provides balance measure of precision and recall of n-gram character, where β determines the relative importance of precision and recall [35], [36], [37]

ROUGE-L measures the longest common subsequence (LCS) between the reference and hypothesis translations. This metric evaluates the fluency and coherence of the translation, reflecting how well the generated translation retains the structure of the reference. The ROUGE-L score is calculated as:

$$\text{Precision} = \frac{\text{LCS(Text Prediction, Text Reference)}}{\text{Words in Text Prediction}} \tag{10}$$

$$\text{Recall} = \frac{\text{LCS(Text Prediction, Text Reference)}}{\text{Words in Text Reference}} \tag{11}$$

$$F \text{ Measure} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{(\text{Precision} + \text{Recall})} \tag{12}$$

V. RESULTS AND ANALYSES

This section presents the results and analyses of the fine-tuning and the proposed custom loss experiments to evaluate the performance improvement over the baseline model. Empirical studies are also conducted on the models to analyze further improvements and to understand the impact of different parameters on the model's effectiveness. Finally, we present user acceptance test and students' post-test to evaluate the overall pipeline SIBI animations system applied to the educational video in the respondents' perspective.

A. FINE-TUNING RESULT

As shown in **TABLE 2**, both models showed significant improvements across all evaluation metrics compared to their baselines. Specifically, the NLLB200 FT model improved by 64.38% in sacreBLEU, 6.62% in chrF++, 18.75% in METEOR, and 10.91% in ROUGE-L over the

TABLE 2. Evaluation results of the fine-tuning experiments on the two pre-trained models, mBART50 and NLLB200, both with and without a custom loss function. For consistency and precision, the values are rounded to two decimal places.

Model	sacreBLEU	chrF++	METEOR	ROUGE L
NLLB200 (Baseline)	16.08	53.86	0.47	0.54
NLLB200 FT	26.25	57.79	0.57	0.61
NLLB200 FT+CL	**27.32**	**59.52**	**0.59**	**0.63**
mBART50 (Baseline)	10.66	36.48	0.31	0.36
mBART50 FT	26.06	58.24	0.57	0.61
mBART50 FT+CL	**26.13**	**59.65**	**0.58**	**0.62**

Note: Baseline is a pretrained model and is not fine-tuned. FT is a fine-tuned model. FT+CL is a fine-tuned model with custom loss. All FT and FT+CL results shown above are the best model for each mode category. NLLB200 FT+CL uses batch size 4, integration type 2, weight 0.2, and penalty 0.1 while mBART50 FT+CL uses batch size 4, integration type 1, weight 0.05, and penalty 0.1.

NLLB200 baseline. Furthermore, adding the custom loss function (NLLB200 FT+CL) resulted in further improvements of 4.09% in sacreBLEU, 2.99% in chrF++, 3.51% in METEOR, and 3.28% in ROUGE-L.

In comparison the mBART50 FT model exhibited improvements of 133.76% in sacreBLEU, 52.16% in chrF++, 67.65% in METEOR, and 60.53% in ROUGE-L over the mBART50 baseline. Additionally, the mBART50 FT+CL model showed further gains of 0.27% in sacreBLEU, 2.42% in chrF++, 1.75% in METEOR, and 1.64% in ROUGE-L.

When examining the results without the custom loss, it is evident that the mBART50 outperformed the NLLB200 in chrF++, METEOR, and ROUGE-L. However, with the custom loss applied, the NLLB200 FT+CL achieved the highest sacreBLEU score of 27.32, compared to 26.13 for the mBART50 FT+CL. This indicates that the custom loss function was more effective in optimizing the performance of the NLLB200 model.

The NLLB200 FT+CL model (with a batch size of 4, using the second weighting variant, weight size of 0.2, and penalty size of 0.1) demonstrated the highest evaluation scores in three out of four metrics compared to its fine-tuned and baseline versions. Meanwhile, the mBART50 FT+CL model (with a batch size of 4, using the first weighting variant, weight size of 0.05, and penalty size of 0.1) also showed improvements, although not as significant as those of the NLLB200 FT+CL. These performance differences highlight the effectiveness of loss function optimization in improving translation quality. The superior performance of the NLLB200 model can be attributed to its training in a larger number of languages, which enables it to better capture linguistic nuances.

Furthermore, to understand the contribution of each parameter setting to the performance of the models, we present an empirical study for each parameter in the following sections. Several categories: batch size, penalty value, and weighting variants, are discussed to provide a comprehensive understanding of their impact on model performance.

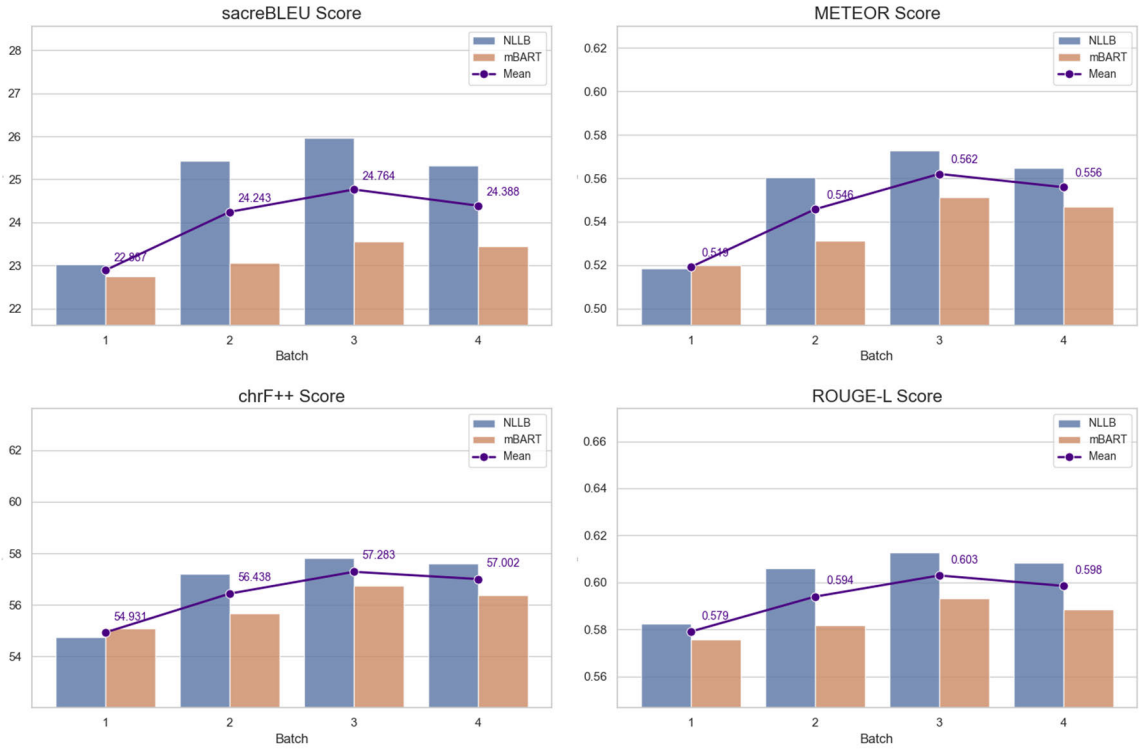
B. EMPIRICAL STUDY ON BATCH SIZE

As previously discussed, four batch sizes were utilized during the fine-tuning process: batch sizes of 1, 2, 3, and 4. **FIGURE 5** illustrates the results of fine-tuning with these various batch sizes. The purpose of the data presentation in the plot is to demonstrate how different batch sizes affect the outcomes of the fine-tuning process. Based on the analysis of the data plot, it can be observed that batch size 3 achieved the best average results with a sacreBLEU score of 24.764, a chrF++ score of 57.283, a METEOR score of 0.562, and a ROUGE-L score of 0.603. These observations indicate that batch size 3 consistently attained the highest average scores across all tested evaluation metrics.

However, it is noteworthy that the best-performing model overall was the NLLB200 model using a batch size of 4, combined with the second weighting variant, a weight size of 0.2, and a penalty size of 0.1. This discrepancy arises because although batch size 3 provided consistently good performance, the best model with batch size 4 outperformed the best model with batch size 3. Thus, while batch size 4 had a lower average performance compared to batch size 3, it still resulted in the highest-performing individual model.

C. EMPIRICAL STUDY ON LOSS INTEGRATION TYPE

Following the fine-tuning process on the base model with the prepared dataset, further optimization efforts were pursued through the implementation of a custom loss function, as outlined in Section III. The implementation of this optimized loss function involved two different weighting variants. The top model variant for each weighting variant, which includes a total of four variants (two models for each variant), is then averaged and summarized into the figure, as depicted in **FIGURE 6**.



AQ:5 **FIGURE 5.** sacreBLEU, chrF++, METEOR, and ROUGE-L evaluation based on batch size.

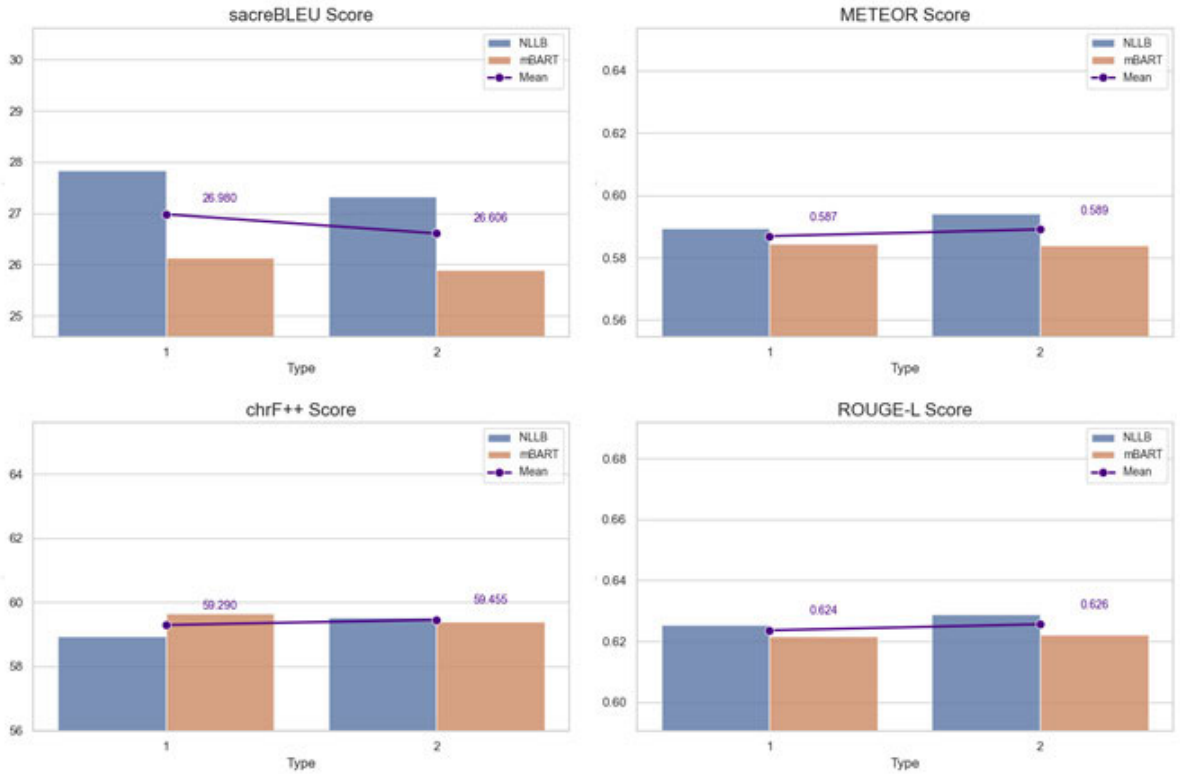


FIGURE 6. sacreBLEU, chrF++, METEOR, and ROUGE-L evaluation based on loss integration variant: custom loss directly added to total loss (integration type 1), and proportional combination of custom and standard loss (integration type 2).

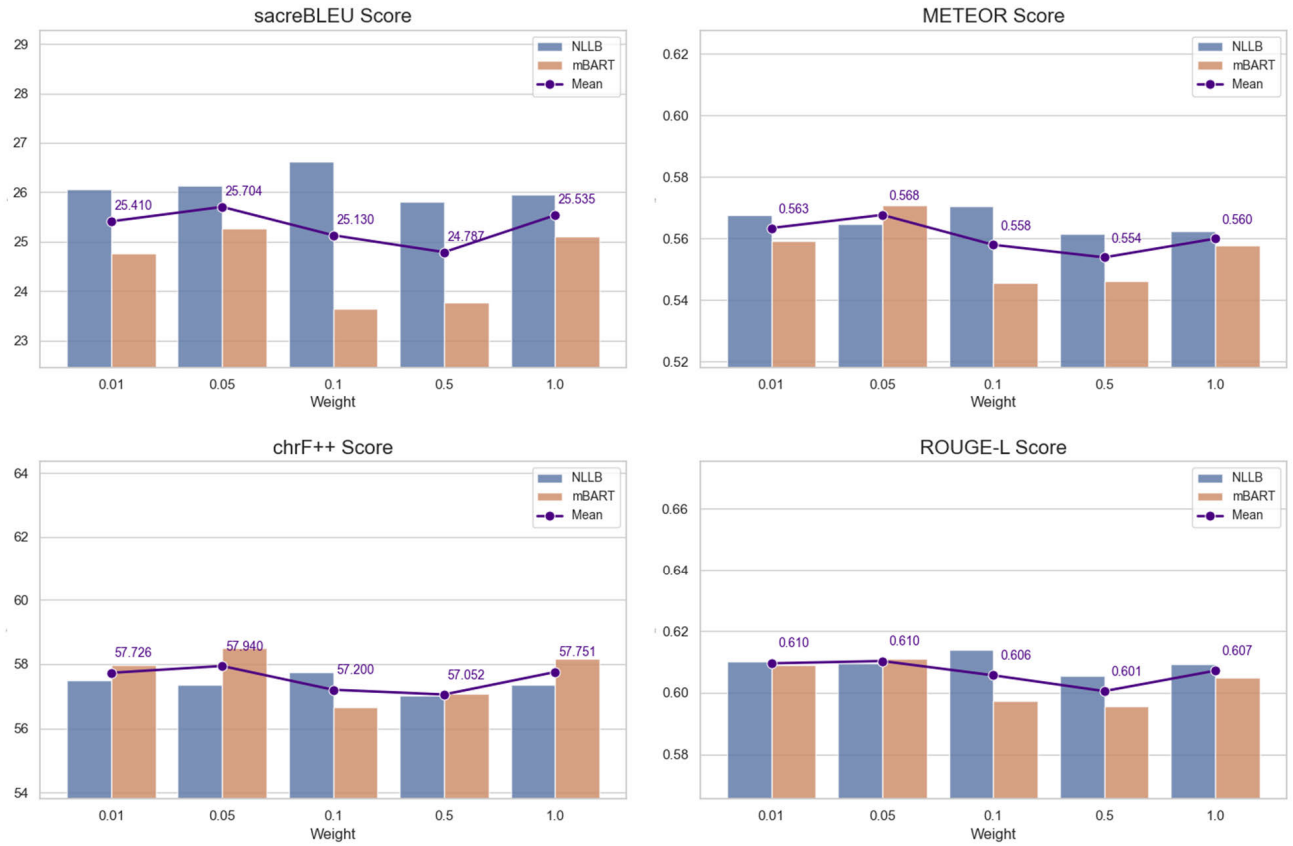


FIGURE 7. sacreBLEU, chrF++, METEOR, and ROUGE-L evaluation based on loss integration type 1 weight value.

As shown in **FIGURE 6**, the second weighting variant demonstrated superior performance in the best model variant when implemented on both models with various parameter configurations compared to the first weighting variant model on three out of four evaluation metrics used. The second variant excelled in the chrF++ metric with a score of 59.51, the METEOR metric with a score of 0.59, and the ROUGE-L metric with a score of 0.62. In contrast, the first variant excelled in the sacreBLEU metric with a score of 27.83.

Based on these results, it can be concluded that, on average, the implementation of the second weighting variant better enhances the model's ability to understand and process language, resulting in more optimal performance compared to the first variant. This conclusion is drawn from the overall performance observed across the various evaluation metrics.

D. EMPIRICAL STUDY ON LOSS INTEGRATION TYPE 1 WEIGHT VALUE

In this section, we examine the influence of different weight values (0.01, 0.05, 0.1, 0.5, and 1) in the loss integration type 1 on the performance of the optimized fine-tuned models. The objective is to identify the best weight configuration that enhances model effectiveness across various evaluation metrics. The evaluation results are averaged and summarized to provide a comprehensive understanding of the performance trends, as shown in **FIGURE 7**.

For the sacreBLEU metric, the NLLB model showed an increase in performance from a weight of 0.01, peaking at a weight of 0.1, then declining at a weight of 0.5, and slightly increasing again at a weight of 1.0. Meanwhile, mBART showed a different pattern, peaking at a weight of 0.05, then experiencing a decline and subsequent increases at higher weights. The peak performance of NLLB at a weight of 0.1 outperformed mBART, with NLLB consistently excelling across all weights. On average, the peak sacreBLEU score was recorded at a weight of 0.05 with a value of 25.704.

In the chrF++ metric, the mBART model outperformed NLLB at almost all weights, peaking at 0.05, while NLLB peaked at 0.1. On average, the highest chrF++ score was recorded at a weight of 0.05 with a value of 57.940.

The METEOR metric showed fluctuations for both models. NLLB peaked at a weight of 0.1, while mBART peaked at 0.05. The peaks for mBART and NLLB were relatively similar but at different weights, with mBART peaking at a weight of 0.05 and NLLB at a weight of 0.1. On average, the peak METEOR score was recorded at a weight of 0.05 with a value of 0.568.

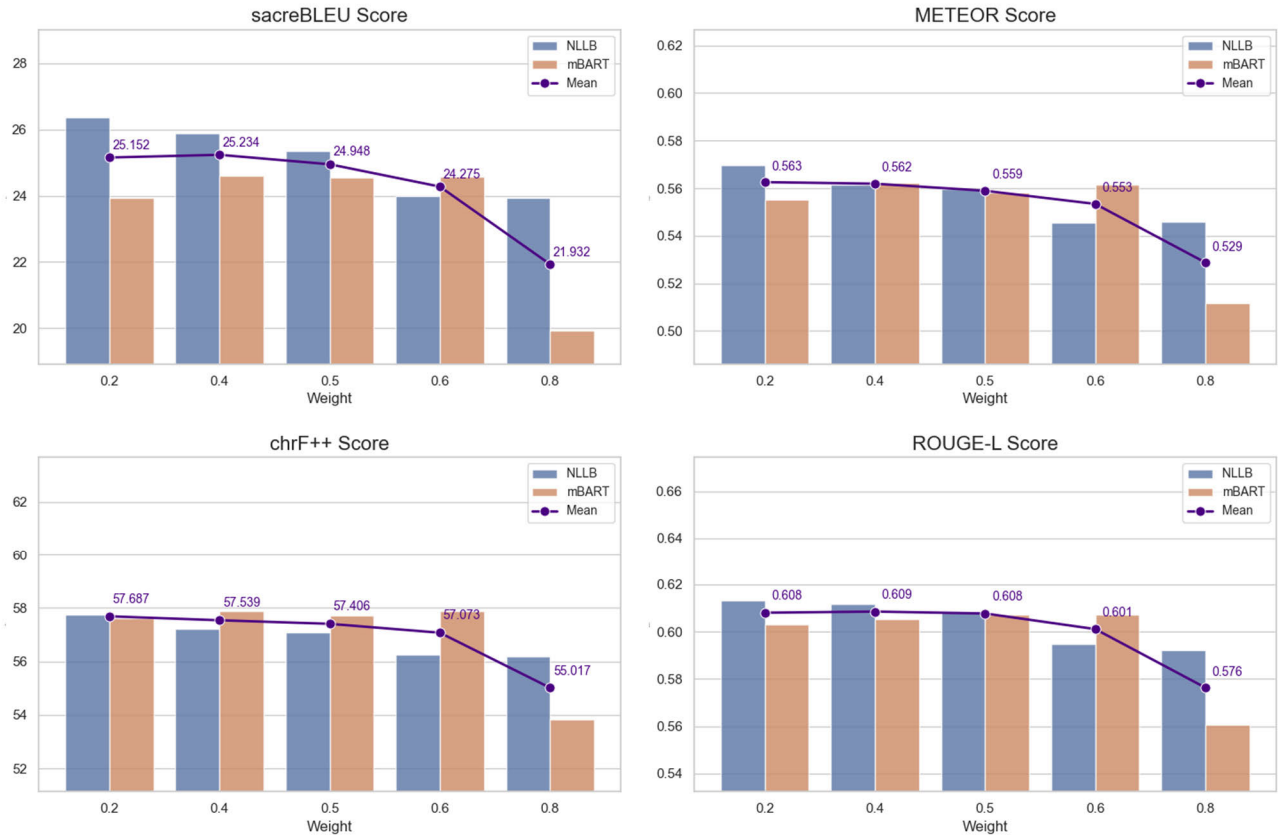


FIGURE 8. sacreBLEU, chrF++, METEOR, and ROUGE-L evaluation based on loss integration type 2 weight value.

In the ROUGE-L metric, both models exhibited noticeable variability. The highest average scores were recorded at weights of 0.01 and 0.05, both with a value of 0.610, while the lowest score was at a weight of 0.5 with a value of 0.601. NLLB peaked at a weight of 0.1, whereas mBART showed a steep decline at this weight. Both models reached their lowest points at a weight of 0.5, with NLLB and mBART showing the same downward trend.

Overall, a weight of 0.05 in the first weighting variant demonstrated the best performance compared to all other weight sizes. This is evidenced by the highest values obtained by this weight in all the evaluation metrics tested. The weight of 0.05 in the first weighting variant achieved a sacreBLEU score of 25.704, a chrF++ score of 57.940, a METEOR score of 0.568, and a ROUGE-L score of 0.610. Therefore, it can be concluded that, in most cases, a weight of 0.05 in the first variant is the optimal choice for achieving the best performance.

E. EMPIRICAL STUDY ON LOSS INTEGRATION TYPE 2 WEIGHT VALUE

This section examines the effect of different weight values (0.2, 0.4, 0.5, 0.6, and 0.8) in the loss integration type 2 variant on model performance. The aim is to identify the weight configuration that maximizes model effectiveness across various metrics. The evaluation results are averaged and summarized to provide a comprehensive understanding of the performance trends, as depicted in **FIGURE 8**.

For the sacreBLEU metric, the NLLB model showed a declining trend with increasing weight values, while the mBART model exhibited an upward trend until a weight of 0.8, where its performance dropped significantly, indicating instability at higher weights. The NLLB model also experienced a performance drop, but it was less severe compared to mBART.

In the chrF++ metric, both models showed a gradual decrease in performance at low to moderate weights, with significant declines at 0.8. NLLB achieved its highest score at a weight of 0.2, while mBART remained consistent but lower. The average score dropped from 57.087 to 55.017 at a weight of 0.8 for both models.

The METEOR metric showed that NLLB consistently outperformed mBART at almost all tested weights, peaking at 0.2 with a score of 0.563. Both models showed declining performance with increasing weights, with the sharpest decrease occurring at a weight of 0.8. The average score dropped from 0.563 at a weight of 0.2 to 0.529 at a weight of 0.8, indicating significant performance degradation.

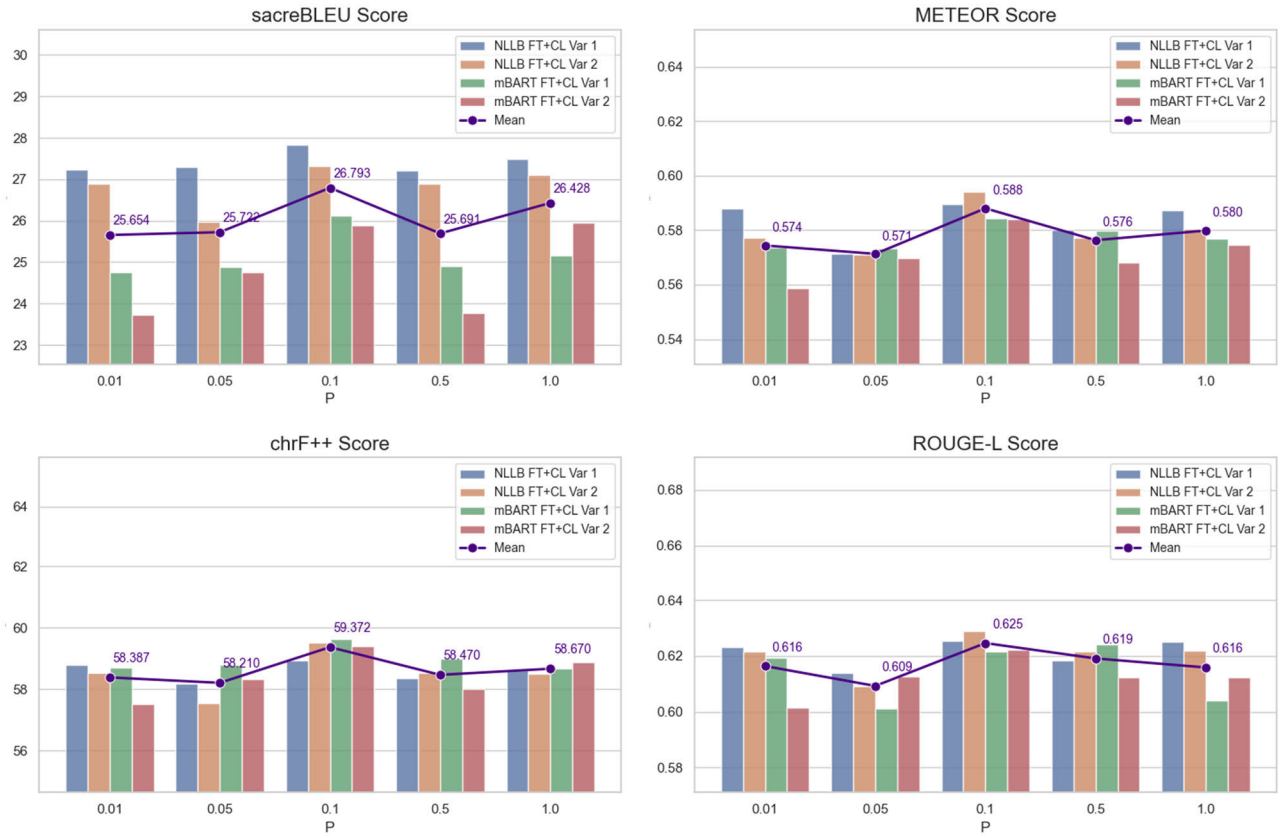


FIGURE 9. sacreBLEU, chrF++, METEOR, and ROUGE-L evaluation based on penalty value.

In the ROUGE-L metric, both models exhibited similar performance patterns to the other metrics, with stable performance at lower weights and declines at higher weights. NLLB's performance gradually decreased, while mBART's performance slightly increased, except at a weight of 0.8. The highest score for this metric was achieved by NLLB at a weight of 0.2. However, at a weight of 0.8, the average score dropped to 0.576, showing that higher weights were not optimal for this metric.

Overall, weights of 0.2 and 0.4 in the second weighting variant demonstrated the best performance. The weight of 0.2 achieved the highest scores in chrF++ (57.687) and METEOR (0.563), slightly outperforming the weight of 0.4. Meanwhile, the weight of 0.4 achieved the highest scores in sacreBLEU (25.234) and ROUGE-L (0.609), slightly outperforming the weight of 0.2. Therefore, it can be concluded that weights of 0.2 and 0.4 are the best choices for achieving the best performance in the second weighting variant.

F. EMPIRICAL STUDY ON PENALTY VALUE

In the final empirical study, we aim to enhance the understanding of factors contributing to the effectiveness of automatic translation models by examining the impact of different penalty values. This experiment utilized the best configurations of optimized models, specifically NLLB FT+CL variants 1 and 2, as well as mBART FT+CL variants 1 and 2. These models had previously demonstrated top performance in evaluation rankings using a penalty of $-log(0.1)$

The success of these configurations in initial testing provided a strong basis for experimenting with various penalty values, namely 0.01, 0.05, 0.1, 0.5, and 1.0, to understand their impact on model performance. By exploring different levels of penalty, this study aimed to identify settings that optimize the trade-off between linguistic precision and the model's adaptability to linguistic variations.

Based on the results of the experiments, it was observed that a penalty value of 0.1, which had previously shown promising results, again yielded superior performance across all evaluation metrics used. Specifically, as shown in **FIGURE 9**, the average scores were 26.793 for sacreBLEU, 59.372 for chrF++, 0.588 for METEOR, and 0.625 for ROUGE-L. These findings highlight the consistent effectiveness of a 0.1 penalty value within the tested range.

From the data analysis, it can be concluded that a penalty value of 0.1 is the most effective choice for enhancing translation performance in both model variants tested. However, it is important to note that this value represents a local optimum

TABLE 3. User acceptance test results of control and experimental group students on learning activities with educational videos.

Question	Control Group	Experimental group
Videos with sign language animations help me understand the material.	2.87	3.33
Videos are too fast.	2.40	2.53
I want to use videos with sign language animations in daily learning activities.	3.00	3.13
The sign language subtitle animations move too fast.	-	2.47
The sign language subtitle animations are too small.	-	2.40

Note: the red font words are only presented for the Experimental Group.

TABLE 4. Post-test results of control and experimental group students on understanding the educational videos.

Post-test	Control Group	Experimental group
Multiple-choice (5)	3.67	3.66
Essay (3)	1.29	2.21
Total (8)	**4.96**	**5.87**

within the scope of the tested penalty values and may not necessarily be the global optimum.

G. USER ACCEPTANCE TEST AND POST TEST

In addition to evaluating the SIBI MT models, our survey team [12] also carried out qualitative and quantitative evaluations of the overall pipeline SIBI Sign Language Subtitle Animations system we proposed. These evaluations involved respondent from senior high SLB students. The evaluations include: user acceptance test and students' post-test. Our survey team divided the students into two groups, control group and experimental group, each consists of 15 students. The control group watched educational videos without SL subtitle animation, while the experimental group used videos with SL animation generated using ground truth SIBI text.

User acceptance testing for students uses a qualitative method by conducting user feedback surveys. In the survey, we asked 3 questions to the control group and 5 questions to the experimental group using a Likert Scale: 1: strongly disagree, 2: disagree, 3: agree, 4: strongly agree. Then, we calculate the average response from all respondents in the group. The UAT results are presented in **TABLE 3**.

The first question feedback showed that students in the experimental group had a higher acceptance rate of 0.46 compared to the control group, who had no experience learning using sign language animation. From the second question feedback, it can be concluded that the speed of the learning video was in accordance with the expectations and comfort of the students in this study. The analysis conducted on the third question shows that students in the experimental group had a slightly higher reuse intention compared to students in the control group.

Student acceptance of sign language animation is also a crucial point that is explored and analyzed, especially the speed factor of sign language animation movements which is the main focus of this study. This is based on the results of the UAT by Permadi [5] study which concluded that students felt that the animation gesture was still too fast. Therefore, we proposed to summarize the input text using our SIBI MT models. From the student feedback from the experimental group for the fourth survey question, an average of 2.46 was obtained on a scale of 1-4. These results indicate that in general respondents do not feel that speed is a problem. Likewise, neither is the size of the animation in the fifth question. Thus, the subtitle animation is in accordance with the expectations and comfort of the students who are respondents in this study.

The post-test quantitative evaluation is used to measure students' understanding of the lesson material in the educational video. Post-test evaluation consists of 5 multiple choice questions (maximum score of 5) and an essay (maximum score of 3) for each video. The testing scenario was carried out by respondents watching videos that corresponded to each control group (without SL animation) and experimental group (with SL animation), and answering the questions afterward. Then it continues with the next video until all the videos have been watched and worked on. The post test results are presented in **TABLE 4**.

Based on these results, we can see that the multiple-choice evaluation scores of the two groups are not much different. However, from the average essay scores we can conclude that students in the experimental group have a better understanding of the video. The essay question format allows students to express their understanding of the video using their own words, therefore it can describe students' understanding well.

VI. CONCLUSION

This research proposes pipeline SIBI sign language subtitle animations that incorporates the SIBI machine translation to solve the problem of too fast-moving gesture animation generated for the educational videos. This research aims to develop an effective machine translation (MT) model for translating Indonesian text into SIBI text that is more concise, has less number of words, and prioritizes the words listed in the SIBI dictionary. We propose transfer learning by fine-tuning the mBART50 and NLLB200 models with a novel custom loss function to improve the translation accuracy and contextual relevance.

The experiments demonstrate that both fine-tuned (FT) and custom loss-enhanced models (FT+CL) significantly outperformed baseline models across all evaluation metrics.

TABLE 5. Models evaluation result.

No		Model	Batch	Integration Type	Weight	Penalty	sacre BLEU	chrF++	METEOR	ROUGE-L	Rank
1		NLLB200 (Baseline)					16.08	53.86	0.48	0.54	1
2		MBART50 (Baseline)					10.66	36.48	0.31	0.37	2
3	Finetuning (FT)	NLLB200 FT	1				23.98	54.62	0.52	0.59	8
4		NLLB200 FT	2				26.01	57.23	0.57	0.62	5
5		**NLLB200 FT**	**3**				**26.25**	**57.79**	**0.57**	**0.61**	**2**
6		NLLB200 FT	4				25.73	57.34	0.57	0.62	4
7		MBART50 FT	1				24.02	56.96	0.54	0.60	6
8		MBART50 FT	2				22.48	57.04	0.54	0.59	7
9		**MBART50 FT**	**3**				**26.06**	**58.24**	**0.58**	**0.62**	**1**
10		MBART50 FT	4				26.07	58.15	0.57	0.61	3
11	NLLB200 Finetuning + SIBIDict Custom Loss (NLLB200 FT+CL)	NLLB200 FT+CL	1	1	0.01	0.1	23.89	55.03	0.53	0.59	80
12		NLLB200 FT+CL	1	1	0.05	0.1	23.81	54.58	0.52	0.58	88
13		NLLB200 FT+CL	1	1	0.10	0.1	23.70	54.80	0.52	0.59	87
14		NLLB200 FT+CL	1	1	0.50	0.1	23.74	54.96	0.53	0.59	85
15		NLLB200 FT+CL	1	1	1.00	0.1	23.01	53.97	0.51	0.58	94
16		NLLB200 FT+CL	1	2	0.20	0.1	23.55	54.73	0.52	0.59	89
17		NLLB200 FT+CL	1	2	0.40	0.1	23.21	54.64	0.52	0.59	91
18		NLLB200 FT+CL	1	2	0.50	0.1	23.58	55.47	0.53	0.59	84
19		NLLB200 FT+CL	1	2	0.60	0.1	24.06	54.36	0.52	0.58	86
20		NLLB200 FT+CL	1	2	0.80	0.1	23.68	56.09	0.53	0.59	83
21		NLLB200 FT+CL	2	1	0.01	0.1	27.26	58.43	0.58	0.61	17
22		NLLB200 FT+CL	2	1	0.05	0.1	26.60	58.10	0.58	0.62	26
23		NLLB200 FT+CL	2	1	0.10	0.1	26.10	56.59	0.56	0.61	52
24		NLLB200 FT+CL	2	1	0.50	0.1	25.63	56.98	0.57	0.61	51
25		NLLB200 FT+CL	2	1	1.00	0.1	27.18	59.17	0.58	0.63	4
26		NLLB200 FT+CL	2	2	0.20	0.1	26.09	55.85	0.55	0.59	70
27		NLLB200 FT+CL	2	2	0.40	0.1	27.38	58.84	0.58	0.63	5
28		NLLB200 FT+CL	2	2	0.50	0.1	24.97	56.58	0.55	0.60	64
29		NLLB200 FT+CL	2	2	0.60	0.1	27.08	58.24	0.57	0.62	28
30		NLLB200 FT+CL	2	2	0.80	0.1	24.99	56.81	0.56	0.60	62
31		NLLB200 FT+CL	3	1	0.01	0.1	26.78	58.33	0.58	0.62	21
32		NLLB200 FT+CL	3	1	0.05	0.1	26.97	58.59	0.59	0.62	12
33		NLLB200 FT+CL	3	1	0.10	0.0	27.24	58.80	0.59	0.62	8
34		NLLB200 FT+CL	3	1	0.10	0.1	27.29	58.19	0.57	0.61	30
35		NLLB200 FT+CL	3	1	0.10	0.1	27.83	58.93	0.59	0.63	2
36		NLLB200 FT+CL	3	1	0.10	0.5	27.21	58.35	0.58	0.62	18
37		NLLB200 FT+CL	3	1	0.10	1.0	27.50	58.63	0.59	0.63	3
38		NLLB200 FT+CL	3	1	0.50	0.1	27.11	58.54	0.59	0.62	9
39		NLLB200 FT+CL	3	1	1.00	0.1	26.36	57.37	0.57	0.61	37
40		NLLB200 FT+CL	3	2	0.20	0.1	27.17	58.89	0.59	0.62	6
41		NLLB200 FT+CL	3	2	0.40	0.1	27.38	58.36	0.58	0.62	11

TABLE 5. *(Continued.)* **Models evaluation result.**

42		NLLB200 FT+CL	3	2	0.50	0.1	26.60	58.10	0.58	0.62	22
43		NLLB200 FT+CL	3	2	0.60	0.1	24.10	56.36	0.56	0.60	69
44		NLLB200 FT+CL	3	2	0.80	0.1	23.73	55.89	0.55	0.59	77
45		NLLB200 FT+CL	4	1	0.01	0.1	26.34	58.16	0.58	0.62	25
46		NLLB200 FT+CL	4	1	0.05	0.1	27.16	58.20	0.57	0.62	29
47		NLLB200 FT+CL	4	1	0.10	0.1	26.10	57.75	0.57	0.61	43
48		NLLB200 FT+CL	4	1	0.50	0.1	26.75	57.62	0.57	0.61	40
49		NLLB200 FT+CL	4	1	1.00	0.1	27.28	58.87	0.58	0.62	7
50		NLLB200 FT+CL	4	2	0.20	0.0	26.89	58.52	0.58	0.62	16
51		NLLB200 FT+CL	4	2	0.20	0.1	25.98	57.54	0.57	0.61	44
52		**NLLB200 FT+CL**	**4**	**2**	**0.20**	**0.1**	**27.32**	**59.52**	**0.59**	**0.63**	**1**
53		NLLB200 FT+CL	4	2	0.20	0.5	26.89	58.52	0.58	0.62	15
54		NLLB200 FT+CL	4	2	0.20	1.0	27.11	58.51	0.58	0.62	14
55		NLLB200 FT+CL	4	2	0.40	0.1	25.55	56.98	0.57	0.61	49
56		NLLB200 FT+CL	4	2	0.50	0.1	26.20	58.21	0.58	0.62	31
57		NLLB200 FT+CL	4	2	0.60	0.1	20.71	56.04	0.53	0.57	90
58		NLLB200 FT+CL	4	2	0.80	0.1	23.35	55.97	0.54	0.59	82
59	MBART50 Finetuning + SIBIDict Custom Loss (MBART50 FT+CL)	MBART50 FT+CL	1	1	0.01	0.1	24.79	57.01	0.55	0.61	60
60		MBART50 FT+CL	1	1	0.05	0.1	25.47	58.09	0.56	0.61	46
61		MBART50 FT+CL	1	1	0.10	0.1	23.92	56.68	0.53	0.60	76
62		MBART50 FT+CL	1	1	0.50	0.1	22.71	55.96	0.51	0.57	92
63		MBART50 FT+CL	1	1	1.00	0.1	24.67	58.14	0.55	0.60	58
64		MBART50 FT+CL	1	2	0.20	0.1	24.40	57.69	0.55	0.59	67
65		MBART50 FT+CL	1	2	0.40	0.1	24.28	56.88	0.55	0.60	72
66		MBART50 FT+CL	1	2	0.50	0.1	23.52	56.24	0.54	0.59	79
67		MBART50 FT+CL	1	2	0.60	0.1	21.30	54.58	0.51	0.58	93
68		MBART50 FT+CL	1	2	0.80	0.1	23.28	56.52	0.53	0.60	78
69		MBART50 FT+CL	2	1	0.01	0.1	25.33	58.49	0.56	0.61	45
70		MBART50 FT+CL	2	1	0.05	0.1	24.96	56.22	0.54	0.59	75
71		MBART50 FT+CL	2	1	0.10	0.1	22.72	56.28	0.54	0.59	81
72		MBART50 FT+CL	2	1	0.50	0.1	24.34	57.17	0.55	0.61	61
73		MBART50 FT+CL	2	1	1.00	0.1	24.20	57.33	0.55	0.60	65
74		MBART50 FT+CL	2	2	0.20	0.1	23.10	57.18	0.56	0.60	74
75		MBART50 FT+CL	2	2	0.40	0.1	25.33	57.70	0.55	0.60	57
76		MBART50 FT+CL	2	2	0.50	0.1	24.62	57.79	0.56	0.61	54
77		MBART50 FT+CL	2	2	0.60	0.1	24.89	58.14	0.56	0.61	48
78		MBART50 FT+CL	2	2	0.80	0.1	24.01	57.70	0.55	0.61	59
79		MBART50 FT+CL	3	1	0.01	0.1	24.98	58.39	0.57	0.61	35
80		MBART50 FT+CL	3	1	0.05	0.1	25.93	58.98	0.58	0.62	19
81		MBART50 FT+CL	3	1	0.10	0.1	24.17	56.62	0.55	0.61	68
82		MBART50 FT+CL	3	1	0.50	0.1	24.17	58.27	0.56	0.60	55

TABLE 5. *(Continued.)* **Models evaluation result.**

83	MBART50 FT+CL	3	1	1.00	0.1	26.22	59.02	0.57	0.61	23
84	MBART50 FT+CL	3	2	0.20	0.1	24.36	58.40	0.57	0.61	41
85	MBART50 FT+CL	3	2	0.40	0.1	23.92	58.41	0.58	0.61	36
86	MBART50 FT+CL	3	2	0.50	0.1	24.65	57.82	0.56	0.61	56
87	MBART50 FT+CL	3	2	0.60	0.0	23.73	57.53	0.56	0.60	66
88	MBART50 FT+CL	3	2	0.60	0.1	24.75	58.32	0.57	0.61	38
89	MBART50 FT+CL	3	2	0.60	0.1	25.89	59.39	0.58	0.62	13
90	MBART50 FT+CL	3	2	0.60	0.5	23.78	58.00	0.57	0.61	50
91	MBART50 FT+CL	3	2	0.60	1.0	25.95	58.88	0.58	0.61	27
92	MBART50 FT+CL	3	2	0.80	0.1	17.57	53.26	0.51	0.55	95
93	MBART50 FT+CL	4	1	0.01	0.1	23.91	57.97	0.56	0.61	53
94	MBART50 FT+CL	4	1	0.05	0.0	24.76	58.70	0.57	0.62	32
95	MBART50 FT+CL	4	1	0.05	0.1	24.88	58.79	0.57	0.60	39
96	**MBART50 FT+CL**	**4**	**1**	**0.05**	**0.1**	**26.13**	**59.65**	**0.58**	**0.62**	**10**
97	MBART50 FT+CL	4	1	0.05	0.5	24.89	59.01	0.58	0.62	20
98	MBART50 FT+CL	4	1	0.05	1.0	25.15	58.67	0.58	0.60	34
99	MBART50 FT+CL	4	1	0.10	0.1	23.75	57.00	0.56	0.60	73
100	MBART50 FT+CL	4	1	0.50	0.1	23.85	56.91	0.56	0.60	63
101	MBART50 FT+CL	4	1	1.00	0.1	25.37	58.13	0.56	0.61	47
102	MBART50 FT+CL	4	2	0.20	0.1	23.86	57.19	0.55	0.60	71
103	MBART50 FT+CL	4	2	0.40	0.1	24.83	58.50	0.57	0.61	42
104	MBART50 FT+CL	4	2	0.50	0.1	25.46	59.05	0.57	0.62	24
105	MBART50 FT+CL	4	2	0.60	0.1	26.24	58.33	0.56	0.61	33
106	MBART50 FT+CL	4	2	0.80	0.1	14.86	47.89	0.45	0.49	96

The NLLB200 FT+CL model achieved the highest overall performance. This highlights the substantial benefits of incorporating custom loss functions. Empirical studies identify the optimal parameters for high performance: a batch size of 3, a weight of 0.05 for the first weighting variant, and a penalty of 0.1. These findings emphasize the importance of precise parameter selection in fine-tuning to achieve optimal results.

From the user acceptance test and students' post-test results it can be concluded that students can understand the material better with the presence of videos equipped with animated sign language subtitles and want to use the animated sign language subtitles in their daily learning activities. In general, students are also satisfied with the speed and the size of the SL animation generated using SIBI text, even though some students still feel that the SL animation moves too fast.

The research faced challenges due to significant differences between annotated texts and transcripts, which hindered the model's ability to learn patterns. The transcripts often lacked punctuation and had poorly segmented sentences. This issue underscores the need for better preprocessing and cleaner data to enhance model performance.

Another key contribution of this research is the creation and cleaning of the SIBIVID-MP12 dataset, containing 441 pairs of Indonesian text and SIBI sign language text, which will be valuable for future research. Additionally, the study introduced a custom loss function with two variants: direct addition to total loss and proportional combination with standard loss. This innovative approach significantly advances the development of MT models for SIBI sign language.

To further improve translation accuracy and contextual relevance, future research should enhance preprocessing methods and use cleaner data. This includes better text normalization, automatic typo detection and correction, and more accurate sentence segmentation. Future works will

include the development of larger and more diverse datasets, supported by ongoing efforts from a broader research team. This will further enhance the model's generalizability and applicability across various domains. Further optimization of parameters using methods like grid search or random search is recommended to find the most effective settings. Integrating measures of token length and semantic similarity into the loss calculation could provide insights into handling sentence structure and maintaining semantic meaning. Future studies should also explore using models with more parameters to capture linguistic nuances and complexities better. Techniques like QLORA [38] could improve efficiency and effectiveness in fine-tuning, reducing computational costs and time without compromising translation quality. A deeper qualitative evaluation involving sign language experts and deaf users in testing and validation will ensure translations are accurate and user appropriate. Exploring ensemble models could further enhance translation accuracy and reliability.

APPENDIX

See Table 5.

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